KITABU QUEST

S5

CORE MATHEMATICS

Mountain Gorilla

Cover scene

students solving mathematics with graphs and models with a mountain gorilla mascot

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE 5

RWA REB-CBC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST S5 Core Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S5 learners study Core Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use models, diagrams, algebra, graphs, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

As you finish each topic, return to the success criteria and correct one answer before moving on.

Trigonometric functions and equations: Lesson Opener

Learning focus: Trigonometric functions and equations

Country link: a Rwanda school, market, farming, building, transport, or data problem for Trigonometric functions and equations.

By the end of this topic, you should be able to:
- solve trigonometric equations, inequalities and related problems using trigonometric functions and equations
- show how to use transformation formulas to simplify the trigonometric expressions
- extend the concepts of trigonometric ratios and their properties to trigonometric equations
- analyse and discuss the solution of trigonometric inequalities
- apply the transformation formulas to simple trigonometric expressions
- use trigonometric functions and equations to model and solve problems involving trigonometry concepts

Inquiry questions:
- How can trigonometric functions and equations help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric functions and equations: Learn

Trigonometric functions and equations teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Trigonometric functions and equations
- Trigonometric
- functions
- equations
- method
- model
- unit
- answer

Trigonometric functions and equations: Worked Example

Worked example for Trigonometric functions and equations: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Trigonometric functions and equations: Vocabulary And Study Skill

Use these words and study moves before you practise Trigonometric functions and equations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Trigonometric functions and equations
- Trigonometric
- functions
- equations
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trigonometric functions and equations without a clear example, method, or evidence.

Trigonometric functions and equations: Guided Activity

Guided activity for Trigonometric functions and equations:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Trigonometric functions and equations: Practice And Correction

Practice for Trigonometric functions and equations:

Main outcome: solve trigonometric equations, inequalities and related problems using trigonometric functions and equations.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to trigonometric functions and equations and solve it.

Trigonometric functions and equations: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trigonometric functions and equations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric functions and equations: Outcome Check

Outcome check for Trigonometric functions and equations: Solve trigonometric equations, inequalities and related problems using trigonometric functions and equations.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric functions and equations: Outcome Check

Outcome check for Trigonometric functions and equations: Show how to use transformation formulas to simplify the trigonometric expressions.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Sequences: Lesson Opener

Learning focus: Sequences

Country link: a Rwanda school, market, farming, building, transport, or data problem for Sequences.

By the end of this topic, you should be able to:
- understand, manipulate and use arithmetic, geometric and harmonic sequences, including convergence
- define a sequence and determine if a given sequence increases or decreases, converges or not
- define and understand arithmetic progressions and their properties
- determine the value of "n", given the sum of the first "n" terms of arithmetic progressions
- show how to apply formulas to determine the "nth" term and the sum of the first "n" terms of arithmetic progressions
- extend the concepts of arithmetic progression to harmonic sequences

Inquiry questions:
- How can sequences help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Sequences: Learn

Sequences teaches how numbers change in a regular way.

Core idea:
- Look at how each term changes to the next term.
- A pattern can add, subtract, multiply, divide, or combine steps.
- Write the rule before extending the pattern.

Key words:
- Sequences
- method
- model
- unit
- answer
- check

Sequences: Worked Example

Worked example for Sequences: Continue the pattern 6, 12, 18, 24, __, __.

1. Compare terms: 12 - 6 = 6, 18 - 12 = 6.
2. Rule: add 6 each time.
3. 24 + 6 = 30, 30 + 6 = 36.

Answer: 30, 36.

Sequences: Vocabulary And Study Skill

Use these words and study moves before you practise Sequences. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Sequences
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Sequences without a clear example, method, or evidence.

Sequences: Guided Activity

Guided activity for Sequences:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Sequences: Practice And Correction

Practice for Sequences:

Main outcome: understand, manipulate and use arithmetic, geometric and harmonic sequences, including convergence.

1. Continue the pattern 5, 10, 15, 20, __, __.
2. Write the rule for the pattern.
3. Create a similar pattern that increases by 4 each time.

Answer check:
Answers: 25, 30. Rule: add 5 each time.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to sequences and solve it.

Sequences: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Sequences.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Sequences: Outcome Check

Outcome check for Sequences: Understand, manipulate and use arithmetic, geometric and harmonic sequences, including convergence.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Sequences: Outcome Check

Outcome check for Sequences: Define a sequence and determine if a given sequence increases or decreases, converges or not.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential equations: Lesson Opener

Learning focus: Logarithmic and exponential equations

Country link: a Rwanda school, market, farming, building, transport, or data problem for Logarithmic and exponential equations.

By the end of this topic, you should be able to:
- solve equations involving logarithms or exponentials and apply them to model and solve related problems
- define logarithm or exponential equations using properties of logarithms in any base
- state and demonstrate properties of logarithms and exponentials
- carry out operations using the change of base of logarithms
- use the properties of logarithms to solve logarithmic and exponential equations
- convert the logarithm to exponential form

Inquiry questions:
- How can logarithmic and exponential equations help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential equations: Learn

Logarithmic and exponential equations teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Logarithmic and exponential equations
- Logarithmic
- exponential
- equations
- method
- model
- unit
- answer

Logarithmic and exponential equations: Worked Example

Worked example for Logarithmic and exponential equations: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Logarithmic and exponential equations: Vocabulary And Study Skill

Use these words and study moves before you practise Logarithmic and exponential equations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Logarithmic and exponential equations
- Logarithmic
- exponential
- equations

- method

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Logarithmic and exponential equations without a clear example, method, or evidence.

Logarithmic and exponential equations: Guided Activity

Guided activity for Logarithmic and exponential equations:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Logarithmic and exponential equations: Practice And Correction

Practice for Logarithmic and exponential equations:

Main outcome: solve equations involving logarithms or exponentials and apply them to model and solve related problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to logarithmic and exponential equations and solve it.

Logarithmic and exponential equations: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Logarithmic and exponential equations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential equations: Outcome Check

Outcome check for Logarithmic and exponential equations: Solve equations involving logarithms or exponentials and apply them to model and solve related problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential equations: Outcome Check

Outcome check for Logarithmic and exponential equations: Define logarithm or exponential equations using properties of logarithms in any base.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Solving equations by numerical method: Lesson Opener

Learning focus: Solving equations by numerical method

Country link: a Rwanda school, market, farming, building, transport, or data problem for Solving equations by numerical method.

By the end of this topic, you should be able to:
- use numerical methods e.g Newton-Raphson method to approximate solution to equations
- illustrate numerical techniques for approximating solutions to equations and be aware of their limitations
- use numerical methods to approximate solutions of equations
- select a numerical method appropriate to a given problem
- derive error estimates for approximate solutions to equations
- appreciate that equations can only be solved approximately using numerical methods

Inquiry questions:
- How can solving equations by numerical method help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Solving equations by numerical method: Learn

Solving equations by numerical method teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Solving equations by numerical method
- Solving
- equations
- numerical
- method
- model
- unit
- answer

Solving equations by numerical method: Worked Example

Worked example for Solving equations by numerical method: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Solving equations by numerical method: Vocabulary And Study Skill

Use these words and study moves before you practise Solving equations by numerical method. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Solving equations by numerical method
- Solving
- equations
- numerical
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Solving equations by numerical method without a clear example, method, or evidence.

Solving equations by numerical method: Guided Activity

Guided activity for Solving equations by numerical method:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Solving equations by numerical method: Practice And Correction

Practice for Solving equations by numerical method:

Main outcome: use numerical methods e.g Newton-Raphson method to approximate solution to equations.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to solving equations by numerical method and solve it.

Solving equations by numerical method: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Solving equations by numerical method.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Solving equations by numerical method: Outcome Check

Outcome check for Solving equations by numerical method: Use numerical methods e.g Newton-Raphson method to approximate solution to equations.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Solving equations by numerical method: Outcome Check

Outcome check for Solving equations by numerical method: Illustrate numerical techniques for approximating solutions to equations and be aware of their limitations.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric and inverse trigonometric functions: Lesson Opener

Learning focus: Trigonometric and inverse trigonometric functions

Country link: a Rwanda school, market, farming, building, transport, or data problem for Trigonometric and inverse trigonometric functions.

By the end of this topic, you should be able to:
- apply theorems of limits and formulas of derivatives to solve problems of refraction of light in a prism, simple harmonic motion problems, and optimisation including trigonometric or inverse trigonometric functions
- extend the concepts of function, domain, range, period, inverse function, limits to trigonometric functions
- extend the concepts of limits and/or differentiation to modeland solve problems involving trigonometric or inverse trigonometric functions
- apply concepts and definition of limits, to calculate the limits of trigonometric functions and remove their indeterminate forms - calculate also their high derivatives
- derive techniques of differentiation to model and solve problems related to trigonometry
- apply technique of differentiation to solve problems involving trigonometric or inverse trigonometric functions such as refraction of light in prism, simple harmonic motion problems,

Inquiry questions:
- How can trigonometric and inverse trigonometric functions help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric and inverse trigonometric functions: Learn

Trigonometric and inverse trigonometric functions teaches how parts of a whole are named and combined.

Core idea:
- The denominator tells how many equal parts make one whole.
- The numerator tells how many of those parts are being used.
- Fractions with the same denominator can be added or subtracted by working with the numerators.
- The denominator stays the same because the size of the parts has not changed.

Key words:
- Trigonometric and inverse trigonometric functions
- Trigonometric
- inverse
- trigonometric
- functions
- method
- model
- unit

Trigonometric and inverse trigonometric functions: Worked Example

Worked example for Trigonometric and inverse trigonometric functions: A learner eats 2/8 of a pineapple in the morning and 3/8 in the afternoon. What fraction is eaten?

1. The denominators are the same, so the parts are equal-sized eighths.
2. Add the numerators: 2 + 3 = 5.
3. Keep the denominator: 8.

Answer: 5/8 of the pineapple was eaten.

Trigonometric and inverse trigonometric functions: Vocabulary And Study Skill

Use these words and study moves before you practise Trigonometric and inverse trigonometric functions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Trigonometric and inverse trigonometric functions
- Trigonometric
- inverse
- trigonometric
- functions
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trigonometric and inverse trigonometric functions without a clear example, method, or evidence.

Trigonometric and inverse trigonometric functions: Guided Activity

Guided activity for Trigonometric and inverse trigonometric functions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Trigonometric and inverse trigonometric functions: Practice And Correction

Practice for Trigonometric and inverse trigonometric functions:

Main outcome: apply theorems of limits and formulas of derivatives to solve problems of refraction of light in a prism, simple harmonic motion problems, and optimisation including trigonometric or inverse trigonometric functions.

1. Draw one whole divided into 8 equal parts. Shade 3/8.
2. Calculate 2/7 + 4/7 and explain why the denominator stays 7.
3. Write one story problem using fractions with the same denominator, then solve it.

Answer check:
Answers: 2/7 + 4/7 = 6/7. The denominator stays 7 because the equal parts are sevenths.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to trigonometric and inverse trigonometric functions and solve it.

Trigonometric and inverse trigonometric functions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trigonometric and inverse trigonometric functions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric and inverse trigonometric functions: Outcome Check

Outcome check for Trigonometric and inverse trigonometric functions: Apply theorems of limits and formulas of derivatives to solve problems of refraction of light in a prism, simple harmonic motion problems, and optimisation including trigonometric or inverse trigonometric functions.

1. Draw 5/8 on a fraction strip.
2. Calculate 3/9 + 4/9.
3. Explain why the denominator does not change.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric and inverse trigonometric functions: Outcome Check

Outcome check for Trigonometric and inverse trigonometric functions: Extend the concepts of function, domain, range, period, inverse function, limits to trigonometric functions.

1. Draw 5/8 on a fraction strip.
2. Calculate 3/9 + 4/9.
3. Explain why the denominator does not change.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector space of real numbers: Lesson Opener

Learning focus: Vector space of real numbers

Country link: a Rwanda school, market, farming, building, transport, or data problem for Vector space of real numbers.

By the end of this topic, you should be able to:
- study linear dependence of vectors of 3 , solve problems related to angles using the scalar product in 3
- define a basis and the dimension of a vector space and give examples of bases of 3
- define addition of vectors of 3 , and multiplication of a vector of R3 by a scalar
- define the dot product and the cross product of two vectors in a three- dimensional vector space and list their properties
- perform operations on vectors in 3 dimensions
- express a vector of 3 as linear combination of given vectors

Inquiry questions:
- How can vector space of real numbers help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector space of real numbers: Learn

Vector space of real numbers teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Vector space of real numbers
- Vector
- space
- real
- numbers
- method
- model
- unit

Vector space of real numbers: Worked Example

Worked example for Vector space of real numbers: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Vector space of real numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Vector space of real numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Vector space of real numbers
- Vector
- space
- real
- numbers
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Vector space of real numbers without a clear example, method, or evidence.

Vector space of real numbers: Guided Activity

Guided activity for Vector space of real numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Vector space of real numbers: Practice And Correction

Practice for Vector space of real numbers:

Main outcome: study linear dependence of vectors of 3 , solve problems related to angles using the scalar product in 3.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to vector space of real numbers and solve it.

Vector space of real numbers: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Vector space of real numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Vector space of real numbers: Outcome Check

Outcome check for Vector space of real numbers: Study linear dependence of vectors of 3 , solve problems related to angles using the scalar product in 3.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector space of real numbers: Outcome Check

Outcome check for Vector space of real numbers: Define a basis and the dimension of a vector space and give examples of bases of 3.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices and determinants of order: Lesson Opener

Learning focus: Matrices and determinants of order

Country link: a Rwanda school, market, farming, building, transport, or data problem for Matrices and determinants of order.

By the end of this topic, you should be able to:
- apply matrix and determinant of order 3 to solve related problems. Demonstrate that a transformation of 3IR is linear and perform operations on linear transformations of 3IR using vectors
- define operations on matrices of order 3
- illustrate the properties of determinants of matrices of order 3
- show that a square matrix of order 3 is invertible or not
- define a linear transformation in 3D by a matrix
- define and perform operations on linear transformations of 3

Inquiry questions:
- How can matrices and determinants of order 3 help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices and determinants of order: Learn

Matrices and determinants of order teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Matrices and determinants of order
- Matrices
- determinants
- order
- method
- model
- unit
- answer

Matrices and determinants of order: Worked Example

Worked example for Matrices and determinants of order: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Matrices and determinants of order: Vocabulary And Study Skill

Use these words and study moves before you practise Matrices and determinants of order. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Matrices and determinants of order
- Matrices
- determinants
- order
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Matrices and determinants of order without a clear example, method, or evidence.

Matrices and determinants of order: Guided Activity

Guided activity for Matrices and determinants of order:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Matrices and determinants of order: Practice And Correction

Practice for Matrices and determinants of order:

Main outcome: apply matrix and determinant of order 3 to solve related problems. Demonstrate that a transformation of 3IR is linear and perform operations on linear transformations of 3IR using vectors.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to matrices and determinants of order and solve it.

Matrices and determinants of order: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Matrices and determinants of order.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices and determinants of order: Outcome Check

Outcome check for Matrices and determinants of order: Apply matrix and determinant of order 3 to solve related problems. Demonstrate that a transformation of 3IR is linear and perform operations on linear transformations of 3IR using vectors.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices and determinants of order: Outcome Check

Outcome check for Matrices and determinants of order: Define operations on matrices of order 3.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Points, straight lines and sphere in 3D: Lesson Opener

Learning focus: Points, straight lines and sphere in 3D

Country link: a Rwanda school, market, farming, building, transport, or data problem for Points, straight lines and sphere in 3D.

By the end of this topic, you should be able to:
- use algebraic representations of points, lines, spheres and planes in 3D space and solve related problems
- define by its coordinates the position of a point in 3D
- define a line using points and direction vector
- define the position vectors of plane
- define the positions of a line and a sphere in 3D
- represent a point and/or a vector in 3D - Calculate the distance between two points in 3D and the mid - point of a segment in 3D

Inquiry questions:
- How can points, straight lines and sphere in 3d help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Points, straight lines and sphere in 3D: Learn

Points, straight lines and sphere in 3D teaches properties of three-dimensional shapes such as cubes, cuboids, and cylinders.

Core idea:
- Solid shapes have faces, edges, and vertices.
- A cuboid has rectangular faces.
- A cube has six equal square faces.
- A cylinder has two circular faces and one curved surface.
- Nets show how flat faces fold to make a solid.

Key words:
- Points, straight lines and sphere in 3D
- Points
- straight
- lines
- sphere
- method
- model
- unit

Points, straight lines and sphere in 3D: Worked Example

Worked example for Points, straight lines and sphere in 3D: Name the faces of a cuboid.

1. Look at the solid shape.
2. A cuboid has 6 faces.
3. Each face is a rectangle.
4. Opposite faces are equal in size.

Answer: A cuboid has 6 rectangular faces, 12 edges, and 8 vertices.

Points, straight lines and sphere in 3D: Vocabulary And Study Skill

Use these words and study moves before you practise Points, straight lines and sphere in 3D. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Points, straight lines and sphere in 3D
- Points
- straight
- lines
- sphere
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Points, straight lines and sphere in 3D without a clear example, method, or evidence.

Points, straight lines and sphere in 3D: Guided Activity

Guided activity for Points, straight lines and sphere in 3D:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Points, straight lines and sphere in 3D: Practice And Correction

Practice for Points, straight lines and sphere in 3D:

Main outcome: use algebraic representations of points, lines, spheres and planes in 3D space and solve related problems.

1. Draw or model a cube, cuboid, and cylinder.
2. Count faces, edges, and vertices where they apply.
3. Sketch a simple net for a cube or cuboid.

Answer check:
Answer check: cube has 6 equal square faces; cuboid has 6 rectangular faces; cylinder has circular faces and a curved surface.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to points, straight lines and sphere in 3d and solve it.

Points, straight lines and sphere in 3D: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Points, straight lines and sphere in 3D.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Points, straight lines and sphere in 3D: Outcome Check

Outcome check for Points, straight lines and sphere in 3D: Use algebraic representations of points, lines, spheres and planes in 3D space and solve related problems.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Points, straight lines and sphere in 3D: Outcome Check

Outcome check for Points, straight lines and sphere in 3D: Define by its coordinates the position of a point in 3D.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Bivariate statistics: Lesson Opener

Learning focus: Bivariate statistics

Country link: a Rwanda school, market, farming, building, transport, or data problem for Bivariate statistics.

By the end of this topic, you should be able to:
- extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines
- define the covariance, coefficient of correlation and regression lines
- analyse, interpret data critically then infer conclusion
- determine the coefficient of correlation, covariance and regression lines of bivariate data of dispersion of a given statistical series
- apply and explain the coefficient of correlation and standard deviation as the more convenient measure of the variability in the

Inquiry questions:
- How can bivariate statistics help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Bivariate statistics: Learn

Bivariate statistics teaches how to collect, organize, read, and explain information.

Core idea:
- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:
- Bivariate statistics
- Bivariate
- statistics
- method
- model
- unit
- answer
- check

Bivariate statistics: Worked Example

Worked example for Bivariate statistics: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Bivariate statistics: Vocabulary And Study Skill

Use these words and study moves before you practise Bivariate statistics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Bivariate statistics
- Bivariate
- statistics
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Bivariate statistics without a clear example, method, or evidence.

Bivariate statistics: Guided Activity

Guided activity for Bivariate statistics:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Bivariate statistics: Practice And Correction

Practice for Bivariate statistics:

Main outcome: extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:
Answer check: totals in the tally table and bar chart should match.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to bivariate statistics and solve it.

Bivariate statistics: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Bivariate statistics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Bivariate statistics: Outcome Check

Outcome check for Bivariate statistics: Extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Bivariate statistics: Outcome Check

Outcome check for Bivariate statistics: Define the covariance, coefficient of correlation and regression lines.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Conditional probability and Bayes theorem: Lesson Opener

Learning focus: Conditional probability and Bayes theorem

Country link: a Rwanda school, market, farming, building, transport, or data problem for Conditional probability and Bayes theorem.

By the end of this topic, you should be able to:
- solve problems using Bayes theorem and use data to make decisions about likelihood and risk
- extend the concept of probability to explain it as a measure of chance
- compute the probability of an event B occurring when event A has already taken place
- interpret data to make decision about likelihood and risk
- apply theorem of probability to calculate the number possible outcomes of occurring independent events under equally likely assumptions
- determine and explain results from an experiment with possible outcomes

Inquiry questions:
- How can conditional probability and bayes theorem help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Conditional probability and Bayes theorem: Learn

Conditional probability and Bayes theorem teaches how likely an event is.

Core idea:
- Some events are certain, likely, unlikely, or impossible.
- A simple experiment can help you compare chances.
- Record results before making a conclusion.

Key words:
- Conditional probability and Bayes theorem
- Conditional
- probability
- Bayes
- theorem
- method
- model
- unit

Conditional probability and Bayes theorem: Worked Example

Worked example for Conditional probability and Bayes theorem: A bag has 4 red counters and 1 blue counter. Which colour is more likely to be picked?

1. Compare the number of counters.
2. Red has 4 chances; blue has 1 chance.
3. More red counters means red is more likely.

Answer: Red is more likely.

Conditional probability and Bayes theorem: Vocabulary And Study Skill

Use these words and study moves before you practise Conditional probability and Bayes theorem. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Conditional probability and Bayes theorem
- Conditional
- probability
- Bayes
- theorem
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Conditional probability and Bayes theorem without a clear example, method, or evidence.

Conditional probability and Bayes theorem: Guided Activity

Guided activity for Conditional probability and Bayes theorem:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Conditional probability and Bayes theorem: Practice And Correction

Practice for Conditional probability and Bayes theorem:

Main outcome: solve problems using Bayes theorem and use data to make decisions about likelihood and risk.

1. Write one certain event, one likely event, and one impossible event.
2. Toss a coin 10 times and record heads/tails.
3. Use your results to say what happened more often.

Answer check:
Answer check: certain means must happen; impossible means cannot happen.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to conditional probability and bayes theorem and solve it.

Conditional probability and Bayes theorem: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Conditional probability and Bayes theorem.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Conditional probability and Bayes theorem: Outcome Check

Outcome check for Conditional probability and Bayes theorem: Solve problems using Bayes theorem and use data to make decisions about likelihood and risk.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Conditional probability and Bayes theorem: Outcome Check

Outcome check for Conditional probability and Bayes theorem: Extend the concept of probability to explain it as a measure of chance.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometric functions and equations: Review Clinic 1

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Trigonometric functions and equations that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Sequences: Review Clinic 2

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Sequences that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Logarithmic and exponential equations: Review Clinic 3

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Logarithmic and exponential equations that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Solving equations by numerical method: Review Clinic 4

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Solving equations by numerical method that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Core Mathematics.

- Trigonometric functions and equations: Explain this term using an example from the book, your school, or your community.
- Sequences: Explain this term using an example from the book, your school, or your community.
- Logarithmic and exponential equations: Explain this term using an example from the book, your school, or your community.
- Solving equations by numerical method: Explain this term using an example from the book, your school, or your community.
- Trigonometric and inverse trigonometric functions: Explain this term using an example from the book, your school, or your community.
- Vector space of real numbers: Explain this term using an example from the book, your school, or your community.
- Matrices and determinants of order: Explain this term using an example from the book, your school, or your community.
- Points, straight lines and sphere in 3D: Explain this term using an example from the book, your school, or your community.
- Bivariate statistics: Explain this term using an example from the book, your school, or your community.
- Conditional probability and Bayes theorem: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Core Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Core Mathematics that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.